

Hardware Sensor Test Documentation: Magnetometer (Compass)

Module: Magnetometer Sensor Testing

Functionality: Directional Accuracy & Magnetic Field Detection

Test Case ID	Description	Pre-conditions	Test Steps	Expected Result	Status
TC_MAG_001	Verify True North and Magnetic North alignment.	Native Compass app is open. Location services (GPS) enabled for calibration.	1. Hold the device flat horizontally. 2. Rotate 360 degrees slowly and cross-reference with a physical analog compass.	The digital compass needle must point to North precisely with an error margin of less than ± 3 degrees.	Pass
TC_MAG_002	Verify real-time raw Microtesla (μT) value changes.	Sensor Test App or Engineering Mode is open.	1. Monitor the total magnetic field intensity (B) values. 2. Move a weak magnetic object close to the top of the phone.	The raw values along X, Y, and Z axes must dynamically fluctuate, reflecting the proximity of the external magnetic source.	Pass
TC_MAG_003	Verify map orientation tracking (Device Heading).	Google Maps or native Maps app is open with active navigation.	1. Start walking in a specific direction. 2. Rotate your body/phone left and right.	The "blue beam/cone" on the map layout must rotate instantly and smoothly, matching the exact physical direction the user is facing.	Pass

Functionality: Calibration Gestures & Interference Handling

Test Case ID	Description	Pre-conditions	Test Steps	Expected Result	Status
TC_MAG_004	Verify calibration prompt triggering and dismissal.	Sensor detects high electro-magnetic interference (Waveform distortion).	1. Place the phone near a strong speaker or heavy metal block. 2. Open the Compass app. 3. Perform the "Figure 8" calibration motion in the air.	1. The app must display a "Calibrate Compass" overlay screen. 2. The overlay must automatically dismiss once the sensor accuracy returns to normal/high.	Pass
TC_MAG_005	Verify magnetic sensor shielding behavior (Metal proximity).	Device is housed inside a third-party magnetic flip cover case.	1. Launch a sensor logging script. 2. Close and open the magnetic latch of the phone case.	The system must handle the sudden magnetic spike without crashing the sensor service daemon or causing permanent calibration offsets.	Pass

Functionality: Advanced Integrations & Low Power State

Test Case ID	Description	Pre-conditions	Test Steps	Expected Result	Status
TC_MAG_006	Verify sensor fusion interaction with Gyro-scope/Accelerometer.	AR Navigation or star-gazing app (e.g., Stellarium) is active.	1. Move the phone rapidly upward toward the sky.	The system must fuse magnetometer data with the gyro to deliver a stable, jitter-free image overlay without lag.	Pass
TC_MAG_007	Verify sensor power gating during Always-On Display (AOD).	Device is locked. Screen is off or AOD is active.	1. Place the device static on a desk. 2. Track power rails via ADB shell dumpsys sensorservice.	The magnetometer's active hardware state must suspend or enter an ultra-low frequency state to optimize background battery performance.	Pass